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Transformer Models in Medical Imaging: A Survey

Summary

This survey reviews transformer models applied to medical imaging. We cover self-attention architectures and their clinical deployment.

INTRODUCTION

Medical imaging has embraced deep learning. Transformers of- fer new capabilities beyond convolutional networks for segmentation tasks.

METHODS

We surveyed 120 papers published between 2020 and 2024. Each paper was annotated for task, architecture and dataset.

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DISCUSSION

Hybrid CNN-transformer models dominate recent benchmarks. Data efficiency remains an open challenge for rare diseases.

CONCLUSION

Transformers are reshaping medical imaging research.

References

[1] X. Chen. TransUNet. 2021.

[2] Y. Liu et al. Swin-UNet. 2022.